

# Vectors: the Geometry of Data

Data becomes arrows; similarity becomes an angle

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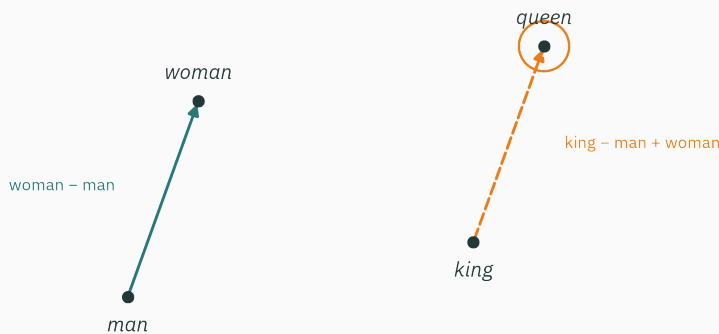
Mathematical Foundations for AI · IIT Gandhinagar

## **Mystery 3: arithmetic on meaning**

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Mystery 1 (the NaN loss) died in L2. Mystery 2 (the moody learning rate) waits for L17. Which leaves the strangest one on the board:

$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$



a 2-D shadow of a 300-dimensional embedding space ( $\rightarrow$  L3)

Subtracting **lists of numbers** somehow captures *gender*. By the end of today this is not magic — it is geometry you can check with a calculator.

## Exhibit B · “people like you”

Every streaming app claims to know “people like you”. But taste isn’t a number — what could “like you” even **mean**, mathematically?

| <b>Film</b>    | <b>You</b> | <b>Priya</b> | <b>Rohan</b> |
|----------------|------------|--------------|--------------|
| Dangal         | 5          | 4            | 0            |
| 3 Idiots       | 4          | 5            | 1            |
| Interstellar   | 1          | 0            | 5            |
| Chak De! India | <b>?</b>   | 5            | 2            |

Who is more “like you” — Priya or Rohan? **By how much?** And should the app pitch you *Chak De!*? Within the hour you’ll compute all three answers — with the same formula that explains king – man + woman.

# Today's one idea

A data point is a **vector**. Similarity between data points is **geometry** — ultimately, an angle.

The route, tool by tool:

| Step   | Tool                | What it unlocks                 |
|--------|---------------------|---------------------------------|
| length | norms $\ x\ $       | how big, how far apart          |
| angle  | dot product, cosine | both mysteries above            |
| shadow | projection          | the germ of least squares & PCA |
| grid   | span, basis         | what coordinates even mean      |

Module 1 starts here: L3 vectors  $\rightarrow$  L4 matrices (they **move** vectors)  $\rightarrow$  L5 eigenvectors  $\rightarrow$  L6 SVD & PCA.

# Learning outcomes

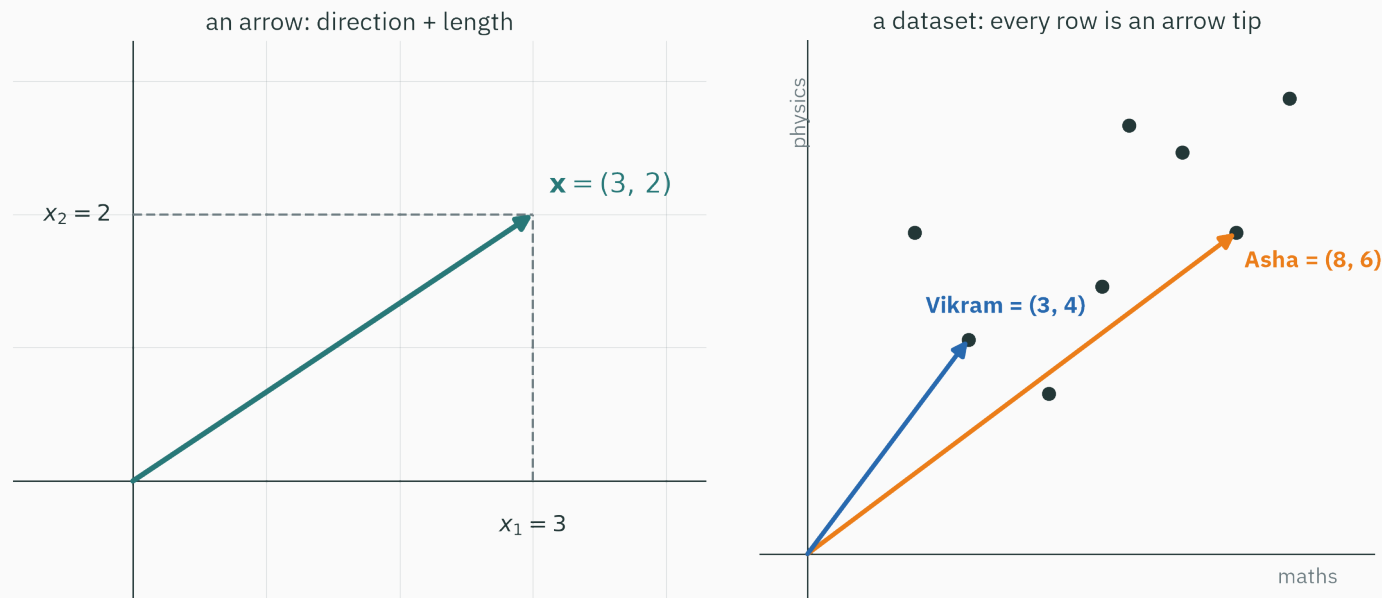
By the end of this lecture you will be able to:

1. Read a vector both ways: an **arrow** in space and a **row** of a dataset.
2. Measure size with **L1 / L2 norms** — and draw each norm’s “unit ball”.
3. Compute the **dot product** three ways: multiply–add, lengths  $\times \cos \theta$ , shadow.
4. Derive  $\cos \theta = x^\top y / (\|x\| \|y\|)$  from the school law of cosines (★).
5. Rank **real word embeddings** by cosine similarity — and dissolve Mystery 3.
6. **Project** one vector onto another and verify the leftover is orthogonal.
7. Say when vectors **span** a line vs the plane, and what an (orthonormal) **basis** buys.

# **Vectors: arrows and data rows**

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# One object, two readings



Left:  $(3, 2)$  is an **arrow** — a direction plus a length. Right: the same kind of pair is a **row of data** — Asha scored  $(8, 6)$  in (maths, physics).

**One object, two readings** — an arrow you can draw, a row you can store. All of today is about switching between them at will.



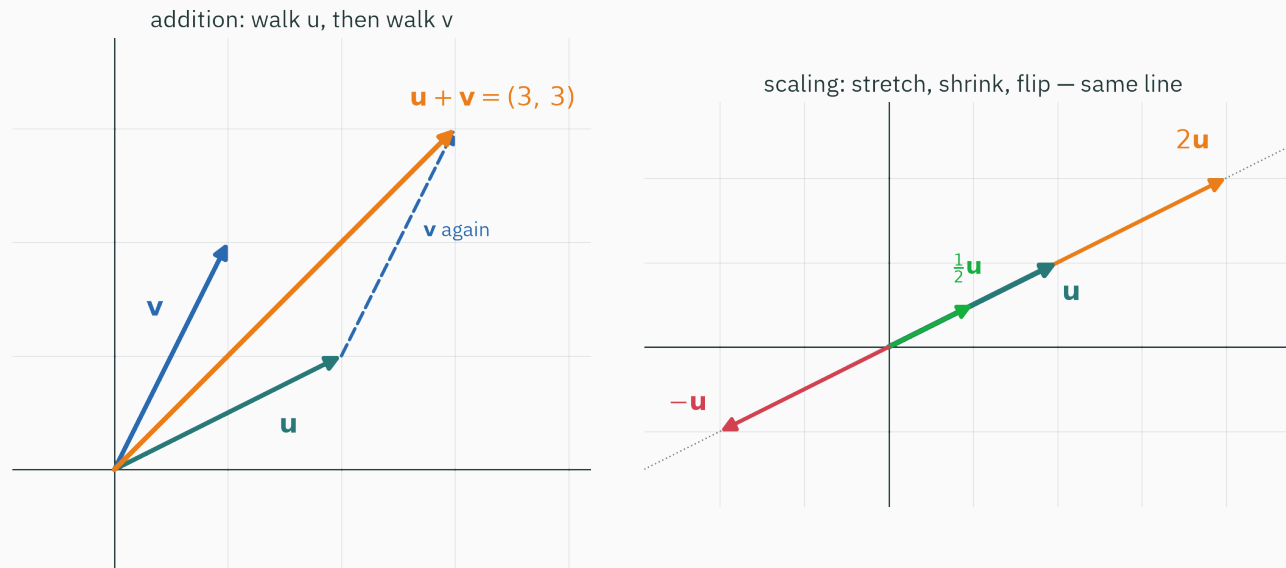
# Everything you meet in AI is one of these

| Object                  | The vector                        | Lives in                      |
|-------------------------|-----------------------------------|-------------------------------|
| Asha's marks            | $(8, 6)$                          | $\mathbb{R}^2$                |
| your film ratings       | $(5, 4, 1, \dots)$                | $\mathbb{R}^{\#\text{films}}$ |
| the word “king” (GloVe) | 50 learned numbers                | $\mathbb{R}^{50}$             |
| one MNIST digit         | $28 \times 28 = 784$ pixel values | $\mathbb{R}^{784}$            |
| 3 s of speech at 16 kHz | 48,000 samples                    | $\mathbb{R}^{48000}$          |

L2 callback: every entry is a float32 — each coordinate lives on last lecture's unevenly spaced number line.

**From here on, “data” means:  $n$  points in  $\mathbb{R}^d$  —  $n$  = how many,  $d$  = how rich.**

# Two moves only: add, and scale



Addition: walk  $u$ , then walk  $v$  — tip to tail. Scaling: stretch, shrink, or flip along the same line.

These two moves are the whole meaning of “linear”. Everything in Module 1 — combinations, span, matrices — is built from exactly these.

# The moves, in numbers and code

With  $u = (2, 1)$ ,  $v = (1, 2)$ : everything happens slot by slot —

$$u + v = (3, 3) \quad 2u = (4, 2) \quad -u = (-2, -1)$$

```
u, v = np.array([2., 1.]), np.array([1., 2.])
u + v      # array([3., 3.])
2 * u      # array([4., 2.])
u * v      # array([2., 2.]) ← slot-by-slot, NOT the dot product
```

$u * v$  multiplies slot-by-slot and keeps a **vector**. The product that carries geometry —  $u @ v$  — collapses two vectors into one number. It gets its own section, two sections from now.

# The notation contract

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} \in \mathbb{R}^d \quad \mathbf{x}^\top = (x_1, \dots, x_d)$$

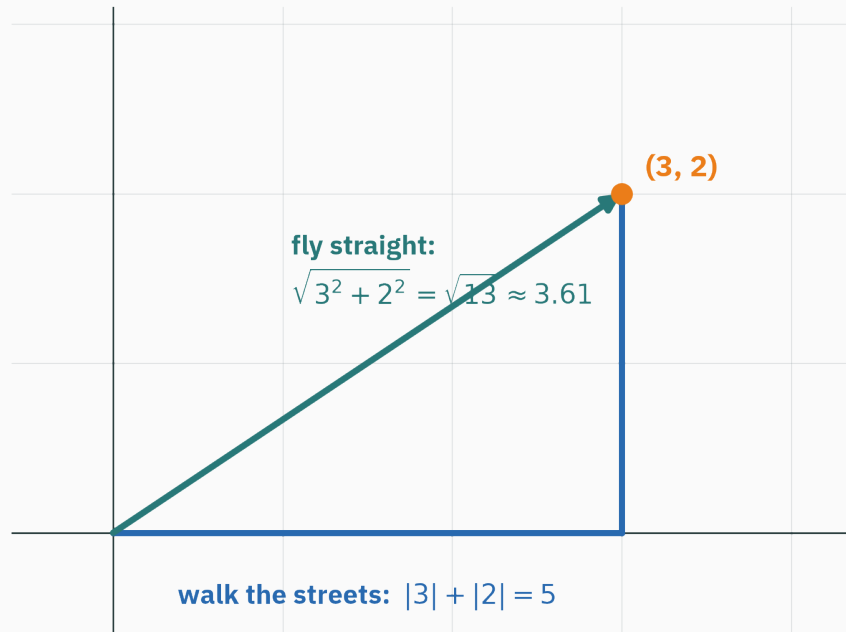
- **bold lowercase**  $\mathbf{x}$ : vectors (columns by default) · *italic*  $x_i$ : scalars & components
- CAPS  $A, X$ : matrices — from L4 on; the dataset is  $X \in \mathbb{R}^{n \times d}$ , one point per row
- $n$  data points,  $d$  dimensions; an unadorned  $\|\mathbf{x}\|$  always means the L2 norm

This is the same contract as MML and the CS229 linear-algebra handout — decks, tutorials, and assignments all speak it. Learn it once, read everything.

# **Norms: how big is a vector?**

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# Two honest ways to measure one trip



Walk the streets:  $|3| + |2| = 5$  blocks. Fly straight:  $\sqrt{3^2 + 2^2} = \sqrt{13} \approx 3.61$ .

Both are legitimate “sizes” of the same vector — they answer different questions. Let’s formalize both.

# The L2 norm · as the crow flies

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2} = \sqrt{x^\top x}$$

For the trip:  $\|(3, 2)\|_2 = \sqrt{13} \approx 3.61$ . Works in any dimension — for  $(1, -2, 2) \in \mathbb{R}^3$ :  
 $\sqrt{1 + 4 + 4} = 3$ .

- This is **the** default:  $\|x\|$  with no subscript means L2.
- It is Pythagoras, applied  $d - 1$  times — flight distance in  $\mathbb{R}^d$ .

Keep the inner form  $\sqrt{x^\top x}$  in view: **the length is secretly a product of  $x$  with itself** — that observation runs the next section.

# The L1 norm · street by street

$$\|x\|_1 = |x_1| + |x_2| + \cdots + |x_d|$$

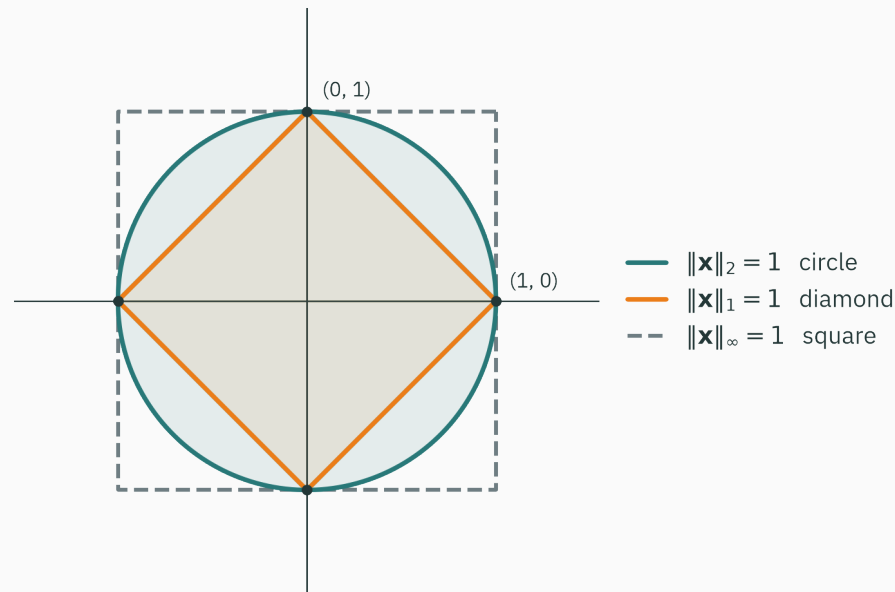
Worked example — a model's errors on four students,  $e = (2, -1, 3, -4)$ :

| Norm           | Value                    | What it judges                          |
|----------------|--------------------------|-----------------------------------------|
| $\ e\ _1$      | $2 + 1 + 3 + 4 = 10$     | total miss — every error counts equally |
| $\ e\ _2$      | $\sqrt{30} \approx 5.48$ | squares amplify — big misses dominate   |
| $\ e\ _\infty$ | $\max = 4$               | only the single worst case              |

One model, three verdicts. **Choosing a norm is choosing what you care about.**



# Every norm draws a different “circle”



All three earn the name “norm” by the same three axioms (MML 3.1): positive unless  $x = 0$ ,  $\|cx\| = |c| \|x\|$ , triangle inequality.

And the L1 diamond has **corners** on the axes — the seed of why L1 regularization zeroes weights out. (ES 335 finishes that story.)

# Unit vectors · direction, distilled

$$\hat{x} = \frac{x}{\|x\|} \quad \text{same direction, length exactly 1}$$

Numbers:  $x = (3, 2)$ :  $\hat{x} = (3, 2)/\sqrt{13} \approx (0.832, 0.555)$  — check:  $0.832^2 + 0.555^2 \approx 1$  ✓

Every nonzero vector splits cleanly:  $x = \|x\| \cdot \hat{x}$  — **how much** × **which way**.

Hold on to this: today's similarity story turns out to be entirely about the **which way** part.

# Distance · a norm in disguise

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Asha (8, 6) vs Vikram (3, 4): difference (5, 2) → flight distance  $\sqrt{29} \approx 5.39$ , street distance 7.

```
asha, vikram = np.array([8., 6.]), np.array([3., 4.])  
np.linalg.norm(asha - vikram)      # 5.385... (L2)  
np.linalg.norm(asha - vikram, 1)   # 7.0      (L1)
```

Every “nearest” in ML — nearest neighbour, nearest cluster centre, nearest embedding — is this one line of code plus a choice of norm.

# One dot product, three readings

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## Reading 1 · multiply matching slots, then add

$$\mathbf{x}^\top \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_d y_d = \sum_{i=1}^d x_i y_i$$

The running pair for the rest of this lecture:  $\mathbf{x} = (3, 2)$ ,  $\mathbf{y} = (4, 2)$ :

$$\mathbf{x}^\top \mathbf{y} = 3 \cdot 4 + 2 \cdot 2 = 16$$

Two vectors in, **one number** out. But what does “16” mean? Hold that thought.

One meaning is already ours — pair  $\mathbf{x}$  with itself:  $\mathbf{x}^\top \mathbf{x} = 9 + 4 = 13 = \|\mathbf{x}\|^2$ .

$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$  — the length was a dot product all along.

## Reading 2 · lengths × alignment

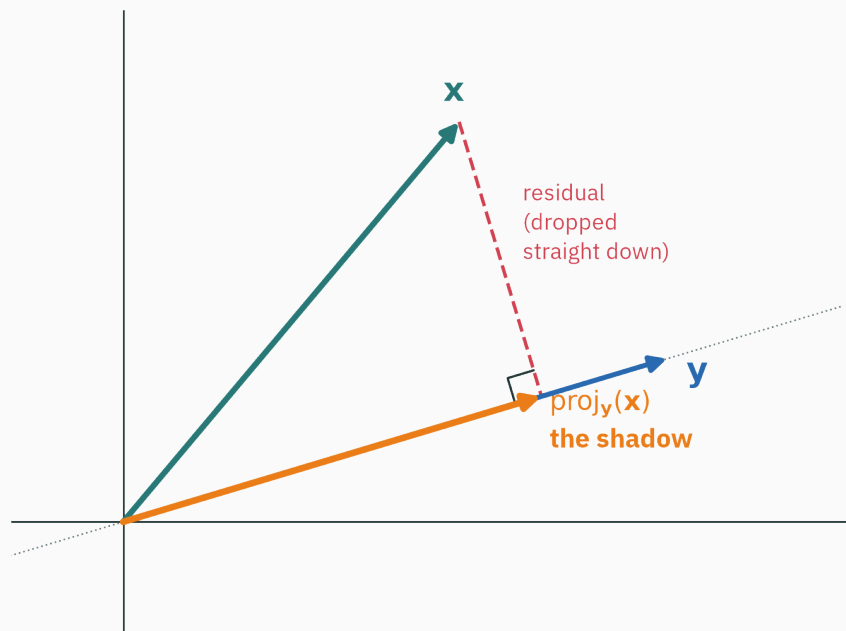
The claim (to be **earned** next section — it is today's ★):

$$x^\top y = \|x\| \|y\| \cos \theta$$

Check it against the running pair:  $\|x\| \|y\| = \sqrt{13}\sqrt{20} \approx 16.12$ , so

$$\cos \theta = \frac{16}{16.12} \approx 0.992 \quad \Rightarrow \quad \theta \approx 7.1^\circ \quad \text{— two nearly parallel arrows.}$$

Pause on how strange this is: pure multiply–add on coordinates apparently **knows the angle**. No protractor was consulted. We will demand proof shortly.



Dot  $x$  against a **unit** direction:  $x^\top \hat{y} = \|x\| \cos \theta =$  the signed **length of  $x$ 's shadow** on  $y$ 's line. Running pair:  $16/\sqrt{20} \approx 3.58$ .

Read it as: “how much of  $x$  points the  $y$  way.”

# The alignment detector

| Angle                          | $\cos \theta$ | $x^\top y$ | Verdict                                          |
|--------------------------------|---------------|------------|--------------------------------------------------|
| acute ( $\theta < 90^\circ$ )  | +             | +          | the vectors agree                                |
| right ( $\theta = 90^\circ$ )  | 0             | 0          | <b>orthogonal</b> — $x \perp y$ , nothing shared |
| obtuse ( $\theta > 90^\circ$ ) | —             | —          | they oppose                                      |

Check:  $(3, 2)^\top (-2, 3) = -6 + 6 = 0 \rightarrow$  perpendicular. The axes too:  $e_1^\top e_2 = 0$ .

Orthogonal directions carry **no information about each other** — the dot product literally sees nothing of one in the other. This idea returns twice today: residuals, and the nicest bases.



# Three readings, one number

```
x, y = np.array([3., 2.]), np.array([4., 2.])
x @ y                                # 16.0      reading 1: multiply-add
nx, ny = np.linalg.norm(x), np.linalg.norm(y)
theta = np.arccos(x @ y / (nx * ny))
np.degrees(theta)                    # 7.125      the angle between them
nx * np.cos(theta)                   # 3.578      reading 3: shadow length
(x @ y) / ny                         # 3.578      same shadow, no trig needed
```

Plant for L4: a matrix multiply is nothing but these, en masse — every entry of  $Wx$  is one dot product  $w_i^\top x$ , one “how much of  $x$  lies along my direction?” question.

$x = (3, 2)$ . Which  $y$  is **orthogonal** to  $x$ ?

**A.**  $y = (2, 3)$

**B.**  $y = (-2, 3)$

**C.**  $y = (3, -2)$

**D.**  $y = (-3, -2)$

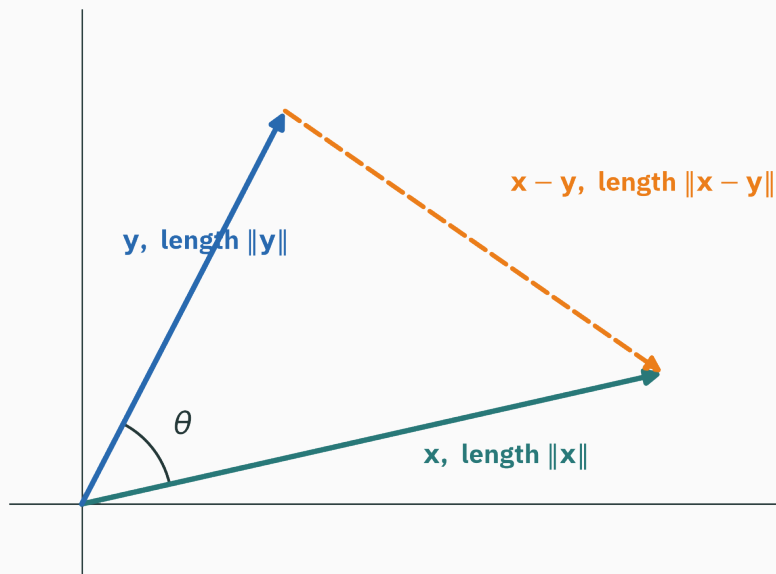
**Correct: B** —  $(-2, 3)$ , since  $3 \cdot (-2) + 2 \cdot 3 = 0$

**Why:** in 2-D, “flip the slots, negate one”:  $(-x_2, x_1)$  is always  $\perp (x_1, x_2)$ . A gives 12 and C gives 5 (acute); D gives  $-13$  — **opposing**, but only exactly zero means perpendicular.

# Why multiply–add measures angle

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**Claim (★):** for any two nonzero vectors,  $x^\top y = \|x\| \|y\| \cos \theta$ .



The two sides of the claim speak different languages — coordinates vs geometry. The bridge: compute the third side  $\|x - y\|^2$  **twice**, once in each language, and compare.

## Step 1 · algebra expands the third side

D DERIVATION

$$\|x - y\|^2 = (x - y)^\top (x - y) \quad (\text{a norm}^2 \text{ is a self-dot})$$

$$= x^\top x - x^\top y - y^\top x + y^\top y \quad (\text{expand the brackets})$$

$$= \|x\|^2 + \|y\|^2 - 2x^\top y \quad (\text{since } y^\top x = x^\top y)$$

Only two facts were used — brackets distribute, and order doesn't matter — and both read straight off the multiply-add formula: check  $x^\top (y + z) = \sum_i x_i (y_i + z_i) = x^\top y + x^\top z$ .

The **law of cosines** (Class 10): any triangle with sides  $a, b, c$  and angle  $\gamma$  opposite  $c$  obeys

$$c^2 = a^2 + b^2 - 2ab \cos \gamma \quad (\text{Pythagoras, plus a tilt correction})$$

Our triangle has sides  $a = \|x\|$ ,  $b = \|y\|$ ,  $c = \|x - y\|$ , and  $\gamma = \theta$ :

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\| \cos \theta$$

Sanity check the correction: at  $\theta = 90^\circ$  the cosine dies and pure Pythagoras remains. ✓

Both computations measured the **same number**  $\|x - y\|^2$ :

$$\|x\|^2 + \|y\|^2 - 2x^\top y = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta$$

Cancel the squared lengths, divide by  $-2$ :

$$x^\top y = \|x\|\|y\|\cos\theta \iff \cos\theta = \frac{x^\top y}{\|x\|\|y\|}$$

Check on the running pair:  $16 = 16.12 \times 0.992 \checkmark$  ( $\theta \approx 7.1^\circ$ , as promised).

Free corollary:  $|\cos\theta| \leq 1$  forces  $|x^\top y| \leq \|x\|\|y\|$  — that's **Cauchy–Schwarz**, bought for nothing.



**Cosine similarity: geometry  
becomes meaning**

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# Angles in $\mathbb{R}^{300}$ , by decree

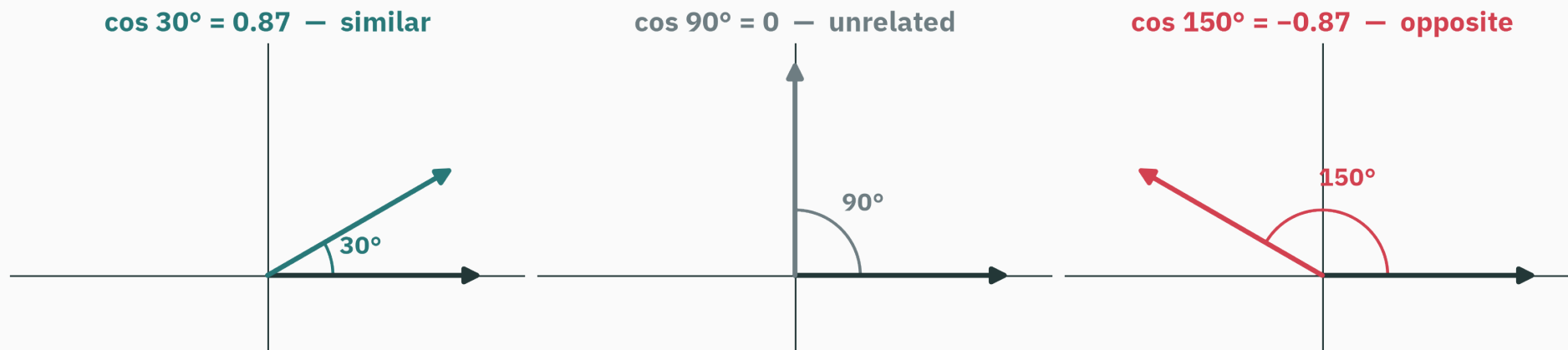
In  $\mathbb{R}^2$  we **saw**  $\theta$  and proved the formula. In  $\mathbb{R}^{300}$  nobody sees anything — so flip the logic and let the formula **define** the angle:

$$\cos_{\text{sim}}(\mathbf{x}, \mathbf{y}) := \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \hat{\mathbf{x}}^\top \hat{\mathbf{y}} \quad \text{— a dot product of unit vectors.}$$

This is legal and safe: Cauchy–Schwarz keeps it inside  $[-1, 1]$ , and any two vectors span an ordinary 2-D plane where it **is** the visible angle.

**Cosine is scale-invariant:**  $\cos_{\text{sim}}(10\mathbf{x}, \mathbf{y}) = \cos_{\text{sim}}(\mathbf{x}, \mathbf{y})$ . **Lengths cancel — only **directions** are compared.**

# The similarity dial, three stops



+1: same story · 0: nothing shared · -1: opposite story. For data, read: **similar / unrelated / opposed**.

## “People like you”, by hand

Ratings on the three films everyone has seen: you = (5, 4, 1), Priya = (4, 5, 0), Rohan = (0, 1, 5).

$$\text{you} \cdot \text{Priya} : \frac{5 \cdot 4 + 4 \cdot 5 + 1 \cdot 0}{\sqrt{42}\sqrt{41}} = \frac{40}{41.5} \approx 0.96 \Rightarrow \theta \approx 15^\circ$$

$$\text{you} \cdot \text{Rohan} : \frac{0 + 4 + 5}{\sqrt{42}\sqrt{26}} = \frac{9}{33.0} \approx 0.27 \Rightarrow \theta \approx 74^\circ$$

**“People like you” = vectors at a small angle to yours. Priya’s 5★ for *Chak De!* becomes your recommendation. Exhibit B: closed.**

# A word is 50 floats

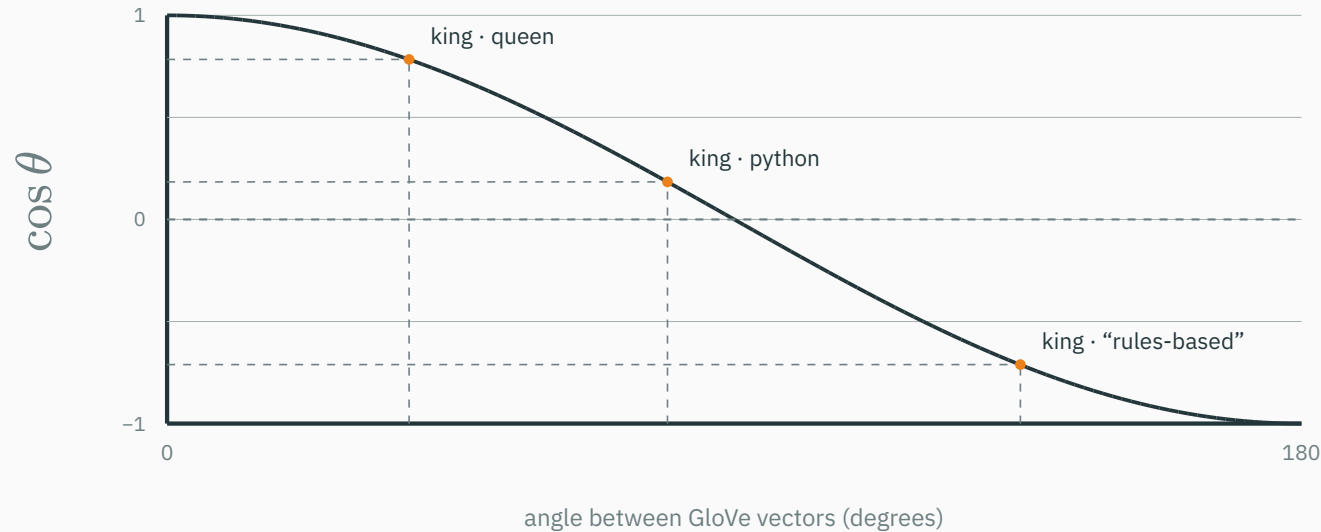
GloVe (2014): read 6 billion words of text; assign every word a vector so that words appearing in **similar contexts** get similar vectors. No dictionary, no grammar — just co-occurrence.

```
import gensim.downloader as api
glove = api.load("glove-wiki-gigaword-50")  # 400k words × 50 floats
glove["king"][:6]
# array([ 0.5045,  0.6861, -0.5952, -0.0228,  0.6005, -0.135 ])
```

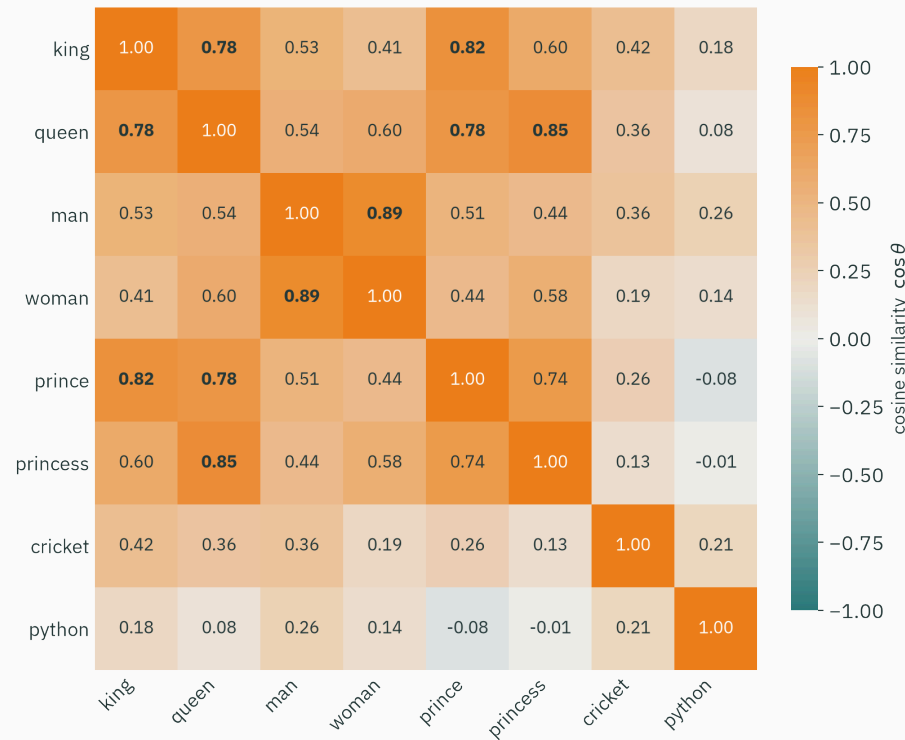
```
def cos_sim(u, v):
    return u @ v / (np.linalg.norm(u) * np.linalg.norm(v))
cos_sim(glove["king"], glove["queen"])  # 0.784 →  $\theta \approx 38.4^\circ$ 
```

Coordinate 17 of “king” means... nothing anyone can name. Individually the 50 float32s (L2!) are gibberish — **the geometry between words carries the meaning.**



More readings: uncle · aunt lands at 40.3°, king · cricket at 65.1°. Semantic neighbours sit at small angles; strangers drift toward 90° and beyond.

# All 28 pairs at once



Blocks are the point: a royal cluster (king–prince 0.82, queen–princess 0.85), a tight man–woman pair (0.89) — and python cold to all, even **opposed** to prince (−0.08).

## Mystery 3 · closed

The bet behind the arithmetic:

$$\mathbf{v}_{\text{king}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}} = \mathbf{v}_{\text{king}} + \underbrace{(\mathbf{v}_{\text{woman}} - \mathbf{v}_{\text{man}})}_{\text{a learned gender shift}}$$

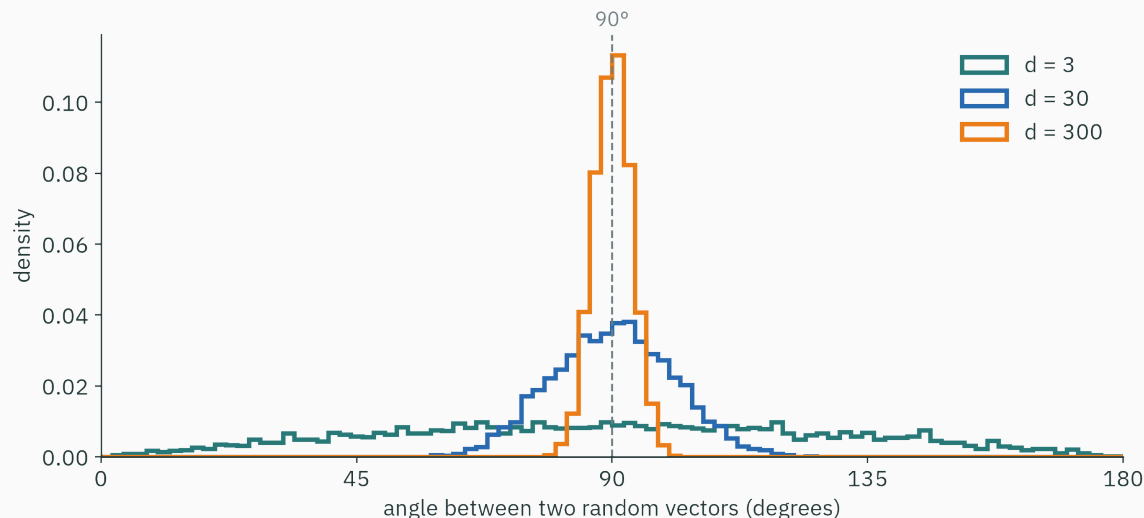
```
glove.most_similar(positive=["king", "woman"], negative=["man"])[0]  
# ('queen', 0.852) ← nearest of all 400k words, by cosine
```

The subtraction never “understood” gender. **Directions in this space consistently encode relationships** — man→woman and king→queen are nearly parallel arrows.

Honest small print: `most_similar` excludes the query words; keep them in and **king itself** outranks queen. The parallelogram is real but approximate — hence “ $\approx$ ”, never “ $=$ ”.



# In high dimensions, strangers are orthogonal



Random pairs in  $\mathbb{R}^3$ : any angle goes. In  $\mathbb{R}^{300}$ : everything piles up at  $90^\circ$  — **unrelated is the default state.**

So king · queen at  $38^\circ$  screams against that background. And the pile-up is good news: high- $d$  space fits astronomically many nearly-perpendicular directions — room for 400k words.

# **Projections: shadows and residuals**

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# From shadow length to shadow vector

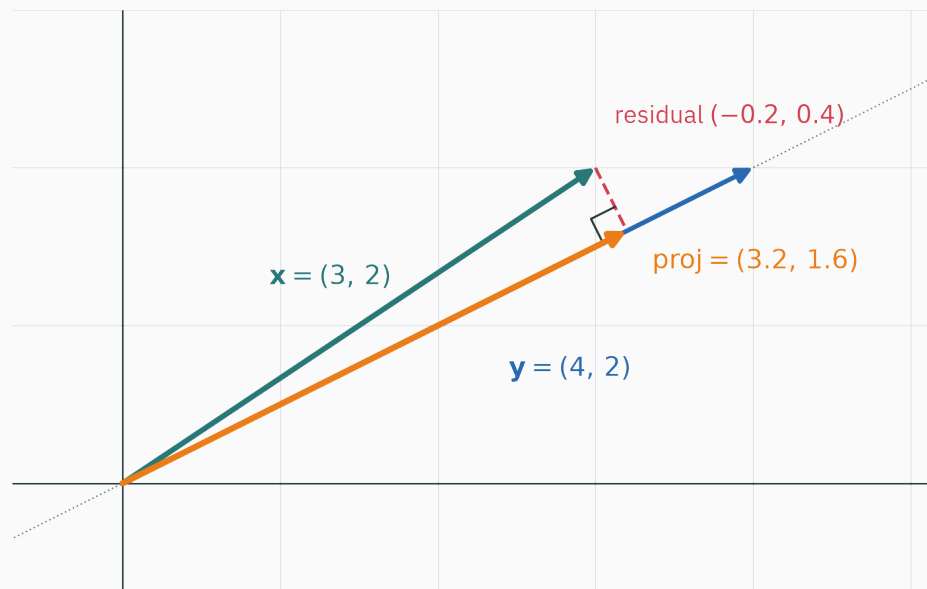
Reading 3 gave a **number** — the shadow's length  $x^\top \hat{y}$ . The shadow itself is that length, aimed along  $\hat{y}$ :

$$\text{proj}_{\mathbf{y}}(\mathbf{x}) = (x^\top \hat{\mathbf{y}}) \hat{\mathbf{y}} = \frac{x^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \mathbf{y}$$

(The right form just substitutes  $\hat{\mathbf{y}} = \mathbf{y}/\|\mathbf{y}\|$  twice and uses  $\|\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{y}$ .)

Read it: **the multiple of  $\mathbf{y}$  that best imitates  $\mathbf{x}$** . Whatever's left — the residual  $\mathbf{x} - \text{proj}$  — is the part of  $\mathbf{x}$  that  $\mathbf{y}$  cannot explain.

# The shadow, in numbers you can check



$$\frac{x^\top y}{y^\top y} = \frac{16}{20} = 0.8 \Rightarrow \text{proj} = 0.8 (4, 2) = (3.2, 1.6)$$

Residual:  $x - \text{proj} = (-0.2, 0.4)$  — and  $(-0.2, 0.4)^\top (4, 2) = -0.8 + 0.8 = 0$ .

Perpendicular, **exactly**.

# Why perpendicular means “closest”

Slide a candidate point along  $y$ 's line and connect it to  $x$ : the connector is shortest where it meets the line at a right angle — any other landing point makes it a hypotenuse over that leg.

$\text{proj}_y(x)$  is the **best approximation of  $x$  on  $y$ 's line** — and “best” is certified by an orthogonal residual.

Widen “ $y$ 's line” to a whole plane of allowed directions and this same right-angle picture becomes **least squares** (L16) and **PCA** (L6). A surprising amount of “learning” is projecting.

# Projection in three lines

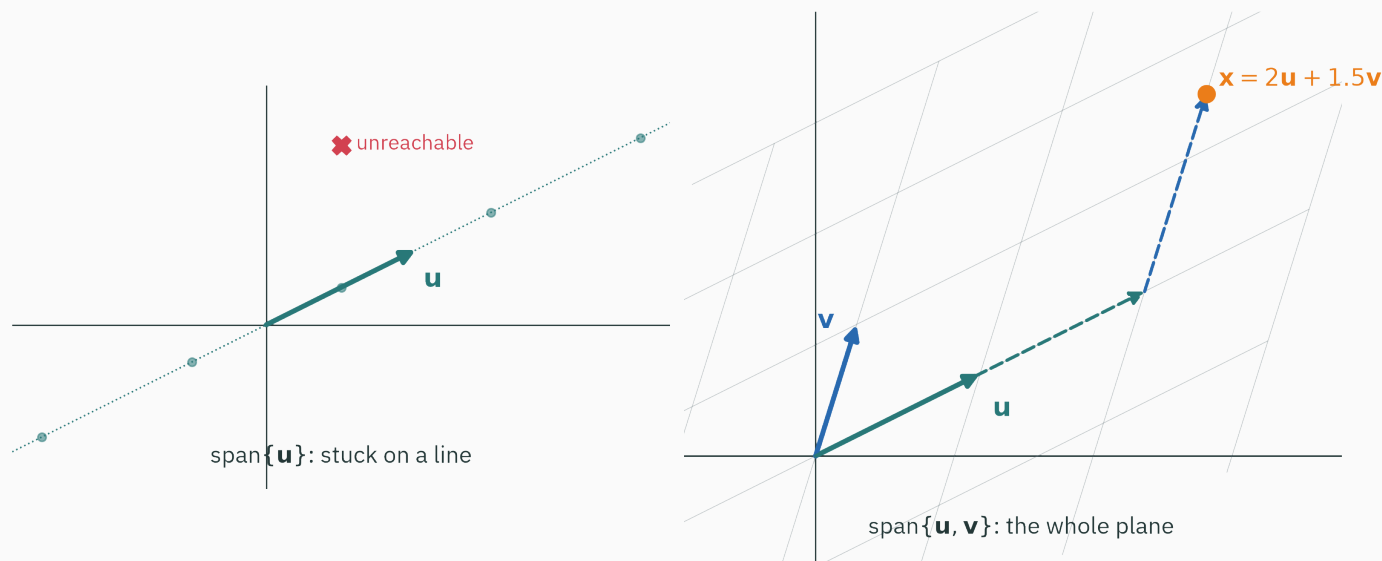
```
x, y = np.array([3., 2.]), np.array([4., 2.])
proj = (x @ y) / (y @ y) * y      # array([3.2, 1.6])
r = x - proj                      # array([-0.2,  0.4])
r @ y                             # 0.0 – the residual is  $\perp$ , always
```

The zero is not luck; it is the definition doing its job. You will reuse exactly these lines inside least squares (L16) and PCA (L6).

**Basis and span: the grid is a choice**

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# Span · where can your moves take you?



$\text{span}\{u\} = \{\alpha u\}$ : scaling alone keeps you on a **line**.  $\text{span}\{u, v\} = \{\alpha u + \beta v\}$ : with two genuinely different directions, walk  $u$ 's then  $v$ 's and **every** point of the plane is reachable.



# When a second vector adds nothing

Take  $u = (2, 1)$  and  $v = (4, 2) = 2u$ :

$\alpha u + \beta v = (\alpha + 2\beta) u$  — every 'combination' is still a multiple of  $u$ .

**Linearly independent** = no vector in the set is a combination of the others.

Dependent = a redundant direction; the span doesn't grow.

Data version: a feature `total = maths + physics` adds a **column** to your table but **no new direction** to the data. That redundancy is why matrices have rank (L4) and why data compresses (L6).

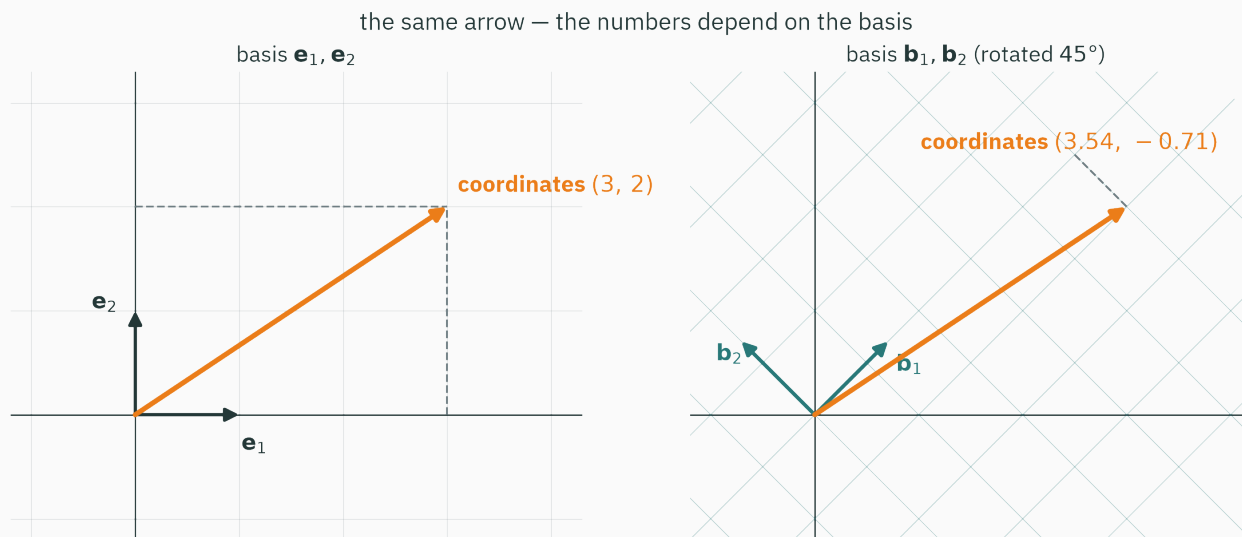
## INTERACTIVE

↗ [textbooks.math.gatech.edu/ila/spans.html](https://textbooks.math.gatech.edu/ila/spans.html)

**Interactive Linear Algebra** (GaTech) — in the span demos, drag  $u$  and  $v$  and watch  $\text{span}\{u, v\}$  collapse from the whole plane to a line the instant the arrows align.

Two minutes of dragging beats twenty static slides — do it tonight. This free textbook is the reference explorable for all of Module 1 (L3–L6).

# A basis is a chosen grid



A **basis** = a set of vectors that is (1) independent and (2) spans the space.  
Consequence: every vector gets **exactly one address** — its coordinates.

Same arrow, two grids, two addresses: (3, 2) vs (3.54, -0.71). **The arrow is real; the numbers are an artifact of the grid you chose.**

# Orthonormal grids read themselves

If  $b_1, b_2$  are **unit length** and **mutually perpendicular** (orthonormal), addresses come free — each coordinate is a projection:

$$x = (x^\top b_1) b_1 + (x^\top b_2) b_2$$

Check against the figure's rotated grid —  $b_1 = (1, 1)/\sqrt{2}$ ,  $b_2 = (-1, 1)/\sqrt{2}$ ,  $x = (3, 2)$ :  
 $x^\top b_1 = 5/\sqrt{2} \approx 3.54$  and  $x^\top b_2 = -1/\sqrt{2} \approx -0.71$  ✓ — the figure's numbers.

**In an orthonormal basis, coordinates are projections: reading an address costs  $d$  dot products.**

L6 teaser: PCA = let the data cloud elect its own favourite orthonormal grid.

**Commit to all four before the reveal.**

**A.**  $\cos \theta$  between  $x$  and  $y$  is 0.8. Your teammate rescales  $x \rightarrow 10x$ . New cosine?

**C.**  $\text{proj}_y(y) = ?$

**B.**  $\text{span}\{(2, 1), (4, 2)\}$ : a point, a line, or the whole plane?

**D.** Can **three** vectors in  $\mathbb{R}^2$  be linearly independent?

- A.** 0.8 — unchanged. The 10 scales  $x^\top y$  and  $\|x\|$  alike and cancels. Cosine sees direction only.
- B.** A line —  $(4, 2) = 2 \cdot (2, 1)$ : the second move is the first in disguise.
- C.**  $y$  itself — the coefficient is  $y^\top y / y^\top y = 1$ : a vector is its own shadow on its own line.
- D.** No — two independent arrows already span  $\mathbb{R}^2$ ; any third is a combination of them. “How many independent directions fit” is exactly what **dimension** means.

# Practice problems

Try on paper; verify in NumPy.

1. For  $x = (1, -2, 2)$ : compute  $\|x\|_1$ ,  $\|x\|_2$ ,  $\|x\|_\infty$ . Which is biggest — and is that always so?
2. A new user rates  $(1, 1, 5)$ . Cosine with you  $= (5, 4, 1)$ ? More or less “like you” than Rohan’s 0.27?
3. Find  $t$  so that  $(3, 2) \perp (4, t)$ .
4. Project  $x = (1, 4)$  onto  $y = (2, 0)$ ; write  $x = \text{shadow} + \text{residual}$  and verify the residual is  $\perp y$ .
5. Show  $(1, 1) \perp (1, -1)$ , normalize both, and write  $(3, 2)$  in this basis via two dot products.
6. In the heatmap,  $\text{king} \cdot \text{cricket} \approx 0.42$  but  $\text{prince} \cdot \text{python} \approx -0.08$ . One-sentence story for each number?

# The reading map

| Source                                       | What it gives you                                               |
|----------------------------------------------|-----------------------------------------------------------------|
| MML Ch 2.1–2.4                               | vectors and vector spaces — today's arrows, done carefully      |
| MML Ch 3.1–3.3                               | norms, inner products, distances — today's rulers               |
| CS229 linear-algebra notes                   | the course's reference handout; keep it at hand through L6      |
| 3b1b <i>Essence of Linear Algebra</i> ch 1–3 | today in moving pictures; chapter 3 is the perfect L4 pre-watch |

All free: [mml-book.github.io](https://mml-book.github.io) · [cs229.stanford.edu/section/cs229-linalg.pdf](https://cs229.stanford.edu/section/cs229-linalg.pdf) · the 3b1b playlist on YouTube. Quiz questions come from lecture beats, and every beat above has a book home.



## Lecture 3 — summary

- **A vector is an arrow AND a data row**; a dataset is  $n$  arrows in  $\mathbb{R}^d$  (each entry a float32 — L2).
- **Norms measure size**: L2 flight, L1 streets,  $L^\infty$  worst case — unit balls: circle, diamond, square.
- **One dot product, three readings**:  $\sum_i x_i y_i = \|x\| \|y\| \cos \theta = \text{shadow} \times \text{length}$ .
- ★ The **law of cosines** turns multiply–add into an angle meter — and **defines** angles in  $\mathbb{R}^{300}$ .
- **Cosine similarity** ranks words, films, people; king – man + woman  $\approx$  queen is parallel displacement arrows, not magic.
- **Projection** = closest point on a line; the residual is orthogonal — the germ of least squares and PCA.
- **Span** is where your moves reach; a **basis** is the grid you narrate coordinates in; orthonormal grids read coordinates by dot products.

Similarity is an angle.

Next: **Matrices as Linear Maps** — the spreadsheet stands up and starts moving: every neural-network layer is a matrix pushing your vectors around.